

BINARY FORMAT SPECIFICATION

The .jack Archive Format

Byte Layout, Endianness, Metadata Dictionary & Bitstream Specification

Version 0.1.0

DRAFT

September 2026

Project	Jackfruit — Basic ZIP-Style File Compression Tool (Huffman Coding)
Team	Saatwik • Tanmayi • Yuvraj • Vibhav
Lead Author	Saatwik (Architecture Owner & Integrator)
Course	UE24CS341A — Software Engineering, PES University
Document Control	FMT-JACK-1.0.0 • Approved Draft for Sprint 1

1. Overview & Objective

The `.jack` file format is a binary container format designed for lossless single-file compression using canonical Huffman coding. The format ensures:

- **Portability** — explicit endianness (Big-Endian / Network Byte Order) for all multi-byte integers.
- **Self-Contained Decompression** — the archive header contains all necessary symbol metadata to reconstruct the exact Huffman decoding tree without external dictionaries.
- **Strict Bit Precision** — a 64-bit valid bit-count prevents trailing byte-padding artifacts.
- **End-to-End Integrity** — a 32-bit CRC checksum guarantees verification of decompressed data integrity.

This specification is the single source of truth that Saatwik's encoder and Tanmayi's decoder are both built against, and that Yuvraj's serializer and Vibhav's validator both read. It is versioned independently of the SRS and SAD (currently Release 1.0.0, "Approved Draft for Sprint 1") because the byte layout, once implemented against, is far more expensive to change than a requirement's wording — any revision here requires re-sign-off from all four roles in the Sign-off Matrix (Section 7), not just the author.

2. Overall File Layout

A `.jack` archive consists of three contiguous sequential sections:

1. **Fixed Header** (18 bytes) — magic bytes, format version, flags, original file size, and CRC-32 checksum.
2. **Metadata & Tree Reconstruction Table** (variable size) — symbol count and frequency dictionary.
3. **Payload Section** (variable size) — 64-bit total payload bit-count followed by the packed Huffman bitstream.

Jackfruit .jack Archive — Overall Byte Layout



Sections are laid out contiguously and sequentially, in this order, with no padding between them.

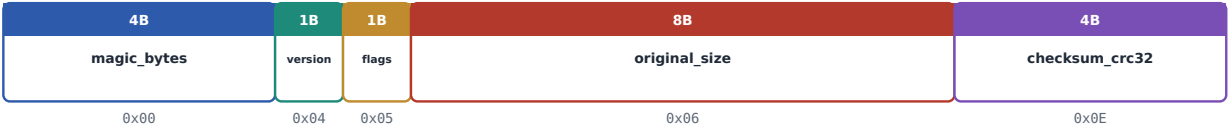
The three sections are ordered deliberately so that a reader (or a parser) never needs to look ahead: the Fixed Header is a constant 18 bytes regardless of content, so it can always be read first without knowing anything else about the file; the Tree Reconstruction Metadata immediately follows and is self-describing via its own `symbol_count` field, so a parser knows exactly how many bytes to consume before reaching the Payload; and the Payload itself begins with its own 8-byte bit-count so that a reader never has to infer where the meaningful bitstream ends and zero-padding begins. This "each section announces the size of what follows" property is what makes `ArchiveReader::readHeader` able to validate structural bounds (Section 6) before a single byte of the potentially large payload section is read into memory.

3. Detailed Byte-Level Layout

All integer fields greater than 1 byte are stored in **Big-Endian (Network Byte Order)**.

Section A: Fixed Header (18 Bytes)

Section A — Fixed Header (18 Bytes, Big-Endian)



Offsets shown are byte offsets from the start of the archive.

Offset	Field Name	Type	Size	Description
0x00	magic_bytes	char[4]	4B	ASCII "JACK" (0x4A, 0x41, 0x43, 0x4B). Identifies valid Jackfruit archives.
0x04	format_version	uint8_t	1B	Format specification version. Currently 0x01 .
0x05	flags	uint8_t	1B	Bitwise flags — Bit 0: IS_EMPTY (1 if original file was 0 bytes); Bit 1: IS_UNCOMPRESSED (fallback if compression expands file); Bits 2–7: Reserved (must be 0).
0x06	original_size	uint64_t	8B	Exact uncompressed file size in bytes (Big-Endian). Supports files up to 2 ⁶⁴ –1 bytes.
0x0E	checksum_crc32	uint32_t	4B	Standard IEEE 802.3 CRC-32 checksum of the original uncompressed file.

Section B: Tree Reconstruction Metadata

To guarantee deterministic Huffman tree reconstruction between Saatwik's encoder and Tanmayi's decoder, the archive stores a sparse frequency table of active symbols.

Offset	Field Name	Type	Size	Description
0x12	symbol_count	uint16_t	2B	Number of unique byte symbols present in the original input (0≤N≤256). If 0, original file was empty.
0x14	symbol_entries	Array	N×9B	Array of N entries, each: symbol (uint8_t, 1B, byte value 0–255) + frequency (uint64_t, 8B, frequency count).

Size of Section B: 2 + (N×9) bytes. Maximum overhead for all 256 byte values is 2 + (256×9) = 2,306 bytes (~2.25 KB).

Section C: Compressed Bitstream Payload

Immediately follows Section B.

Offset	Field Name	Type	Size	Description
Variable	total_bits	uint64_t	8B	Exact number of valid encoded bits in the compressed payload stream.
Variable	bitstream_data	uint8_t[]	M bytes	Encoded Huffman bitstream packed into bytes. M = [total_bits / 8].

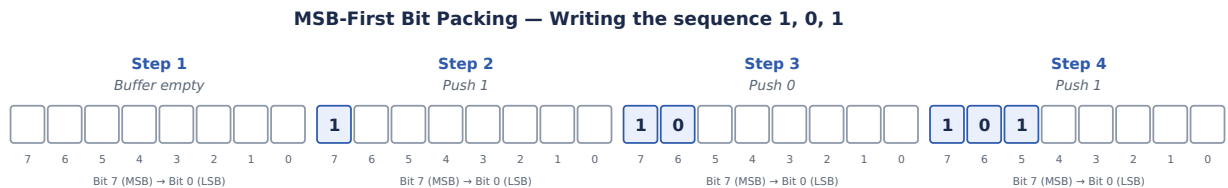
4. Shared Bit-Ordering Convention (BitWriter & BitReader)

CRITICAL CONTRACT — SAATWIK & TANMAYI

All bits are written and read **MSB-First** (Most Significant Bit First). Any mismatch between the two implementations makes every archive silently undecodable.

Packing Convention (BitWriter — Saatwik)

Accumulate bits in an internal 8-bit buffer starting from the Most Significant Bit (Bit 7) down to the Least Significant Bit (Bit 0).



- When 8 bits accumulate, the byte is written to the output stream, and the buffer is cleared.
- Upon calling `flush()`, if any bits remain ($1 \leq \text{bits} \leq 7$), the remaining lower bits in the byte are padded with 0s, and the final byte is emitted.

Unpacking Convention (BitReader — Tanmayi)

- Read bytes sequentially from the stream.
- Extract bits from the current byte from Bit 7 down to Bit 0 using: `bit = (byte >> (7 - bit_index)) & 1`
- Terminate reading exactly when `read_bits_count == total_bits`. Do not read padding bits in the final byte.

5. Edge Case Handling

5.1 Empty File (0 Bytes)

- `original_size = 0`
- `flags = 0x01 ( IS_EMPTY )`
- `checksum_crc32 = 0x00000000`
- `symbol_count = 0`
- `total_bits = 0`
- Total archive size: exactly **28 bytes**.

5.2 Single Unique Symbol (e.g. "AAAAAA")

- `symbol_count = 1`.
- By definition, a Huffman tree requires at least two nodes. For single-symbol files, assign the trivial code `0` of length 1 bit to the single symbol.
- `total_bits = original_size`.

5.3 All 256 Bytes Present

- `symbol_count = 256`. Handled cleanly by `uint16_t`.

6. Validation & Integrity Requirements (Yuvraj & Vibhav)

The five checks below are listed in the exact order they must be performed, and that order is itself a security property, not a stylistic choice: each check is strictly cheaper than the one after it and rejects a wider class of malformed input earlier, so that an attacker-controlled file is discarded before the system does any work proportional to its (possibly falsified) declared size. Checking the CRC-32 first, for example, would be both wasteful (it requires decompressing first) and unsafe (decompression of an unvalidated header is exactly the operation Sections 3.9 and 5.1 of the SRS/SAD identify as the Denial-of-Service risk).

When parsing a `.jack` archive, `ArchiveReader` and `IntegrityValidator` must enforce, in order:

1. **Magic Bytes Check** — if the first 4 bytes $\neq$ "JACK", immediately abort with `ERROR_INVALID_MAGIC`.
2. **Version Check** — if `format_version > 1`, abort with `ERROR_UNSUPPORTED_VERSION`.
3. **Symbol Count Bounds** — if `symbol_count > 256`, abort with `ERROR_CORRUPT_HEADER`.
4. **Payload Length Check** — file size must equal $18 + 2 + (N \times 9) + 8 + \lceil \text{total_bits} / 8 \rceil$. If the physical file is smaller, abort with `ERROR_TRUNCATED_ARCHIVE`.
5. **CRC Check** — after decompression, compute the CRC-32 of the restored bytes. If calculated CRC $\neq$ `checksum_crc32`, report `CORRUPT_ARCHIVE_CHECKSUM_MISMATCH`.

7. Worked Example: Compressing the String "ABRACADABRA"

This section traces a complete, concrete example through the format defined above, for readers who find byte-offset tables easier to verify against an actual worked case. The 11-byte ASCII input `ABRACADABRA` has the following symbol frequencies: A×5, B×2, R×2, C×1, D×1 (5 unique symbols).

7.1 Header Fields

Field	Value for this example
magic_bytes	4A 41 43 4B ("JACK")
format_version	01
flags	00 (not empty, not stored-uncompressed)
original_size	00 00 00 00 00 00 00 0B (11 bytes, Big-Endian)
checksum_crc32	CRC-32 of the 11 ASCII bytes <code>ABRACADABRA</code> (computed at encode time)
symbol_count	00 05 (5 unique symbols)

7.2 Example Canonical Codes

One valid canonical assignment (shortest codes to the most frequent symbol, ties broken by lowest byte value, per JACK-F-002) is:

Symbol	Frequency	Example Code	Length
A	5	0	1 bit
B	2	100	3 bits
R	2	101	3 bits
C	1	110	3 bits
D	1	111	3 bits

Encoding `A B R A C A D A B R A` with this table yields the bit sequence `0 100 101 0 110 0 111 0 100 101 0`, which is 23 bits total — so `total_bits = 23`, packed MSB-first into `M = ⌈23/8⌉ = 3` payload bytes (the last byte padded with 1 trailing zero bit). The original 11 bytes (88 bits) are represented in 23 encoded bits plus the fixed 18-byte header and small metadata table — illustrating why Jackfruit's compression ratio is most favourable on larger, more repetitive files rather than short strings like this illustrative example, where header overhead dominates.

7.3 Resulting Archive Size

18 (header) + 2 (symbol_count) + 5×9 (symbol_entries) + 8 (total_bits) + 3 (bitstream_data) = **76 bytes** for an 11-byte input — a net expansion, which is expected and acceptable for this illustrative short string; the throughput and ratio requirements in the SRS (JACK-NF-001) are measured against realistic multi-kilobyte files, not pathological short inputs like this one.

8. Sign-off Matrix

Role	Member	Status	Notes
Architecture & Compression Lead	Saatwik	Author / Approved	Encoder & BitWriter designed against this spec.
Archive & Statistics Lead	Yuvraj	Co-Author / Pending Review	Serializer & parser logic mirrors this layout.
Testing & Decompression Lead	Tanmayi	Pending Review	Confirmed MSB-first BitReader agreement.
Security & Integrity Lead	Vibhav	Pending Review	Confirmed CRC-32 field position & bounds checks.

— End of Jackfruit Archive Binary Format Specification v1.0.0 —